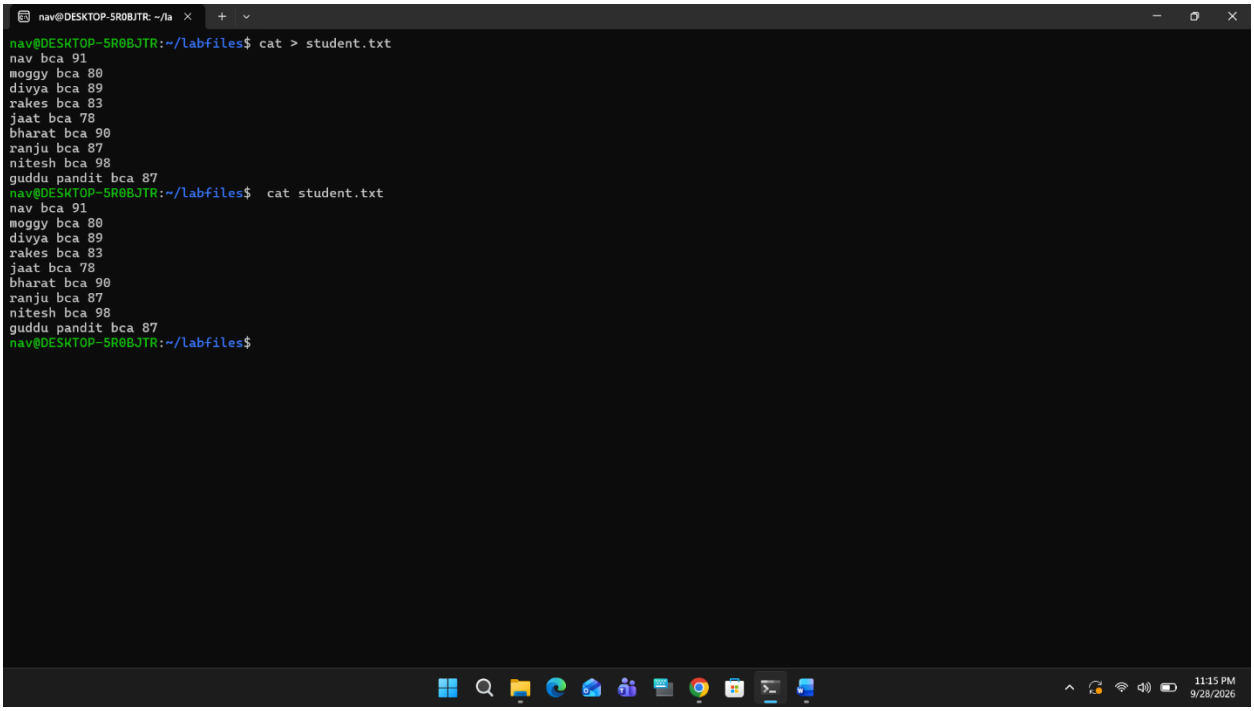
**LINUX LABORATORY**

## Question 2

**Q.2. Demonstrate pipes & redirection (>, >>, <, 2>, 2>>, 2>&1, |) with cat, grep, sort, uniq, wc, tee to generate filtered reports.**

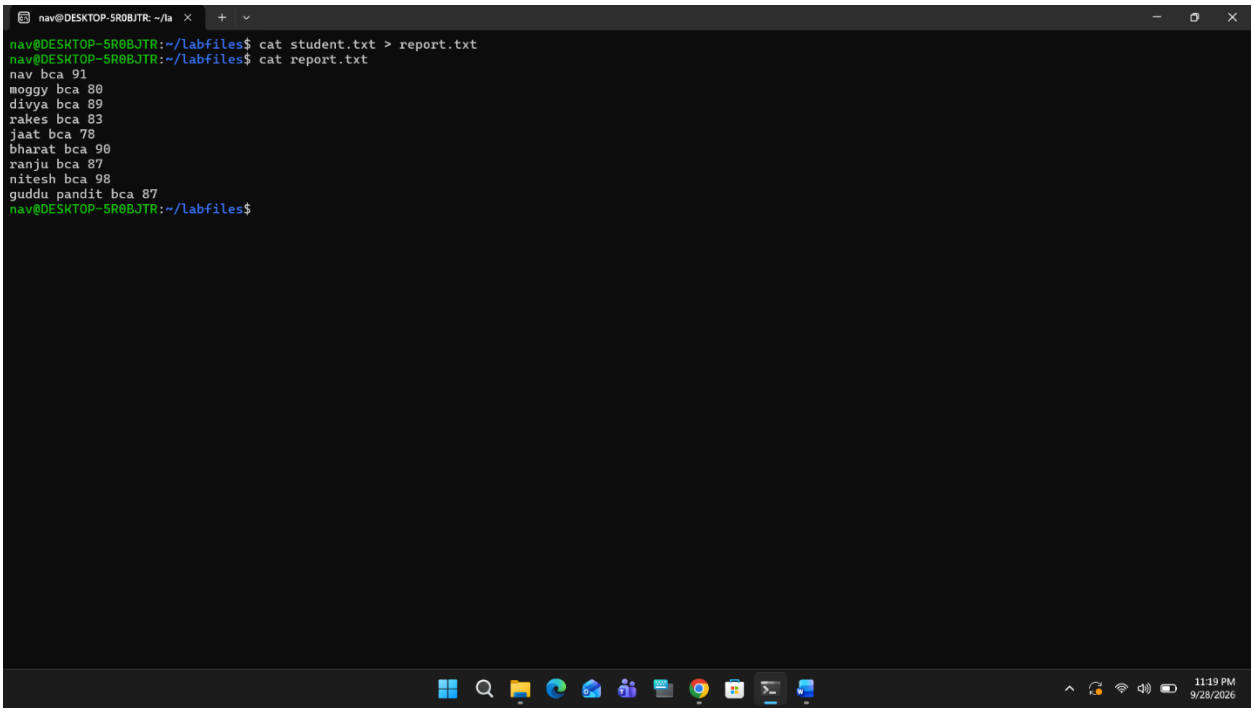
### Step 1 – Create and display the student data

We created students.txt using cat and entered the student records. The file contents were then displayed to verify the data.



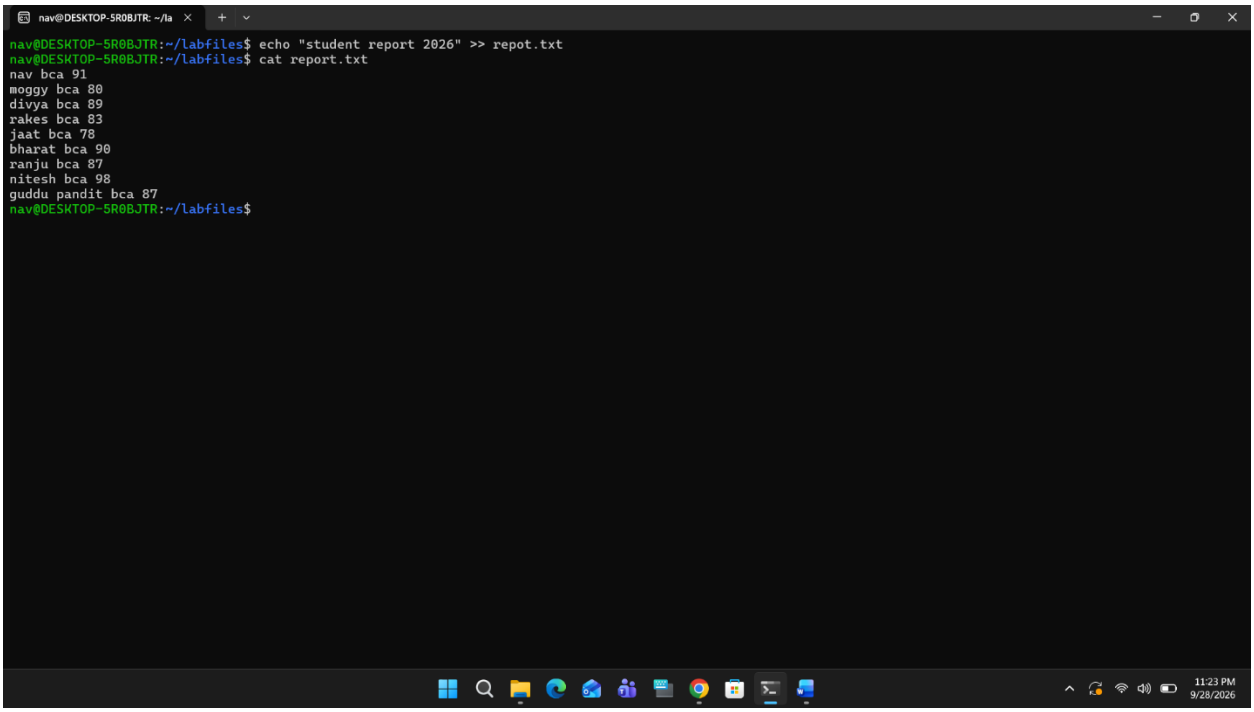
**Step 2 – Redirect standard output using >**

The > operator redirected the output of cat into report.txt. The report file was then displayed to verify that the contents were written successfully.



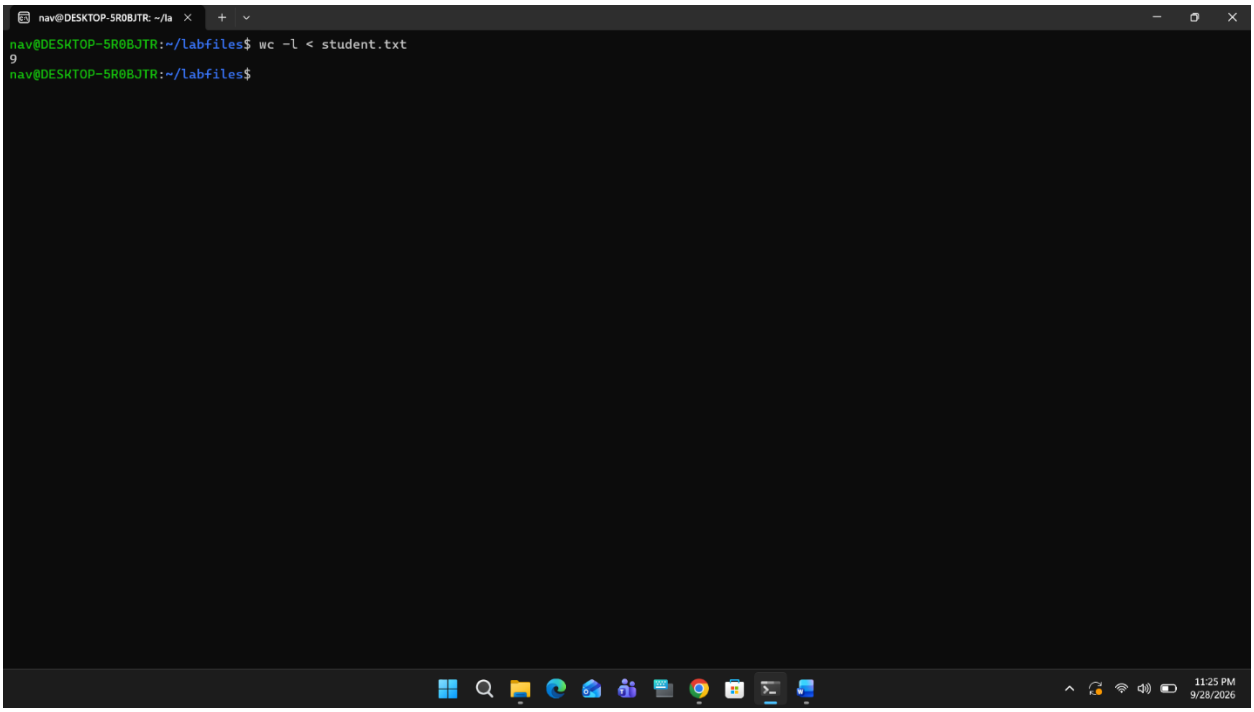
**Step 3 – Append output using >>**

The >> operator appended new text to the existing report.txt file without overwriting its previous contents. The file was displayed again to verify the append operation.



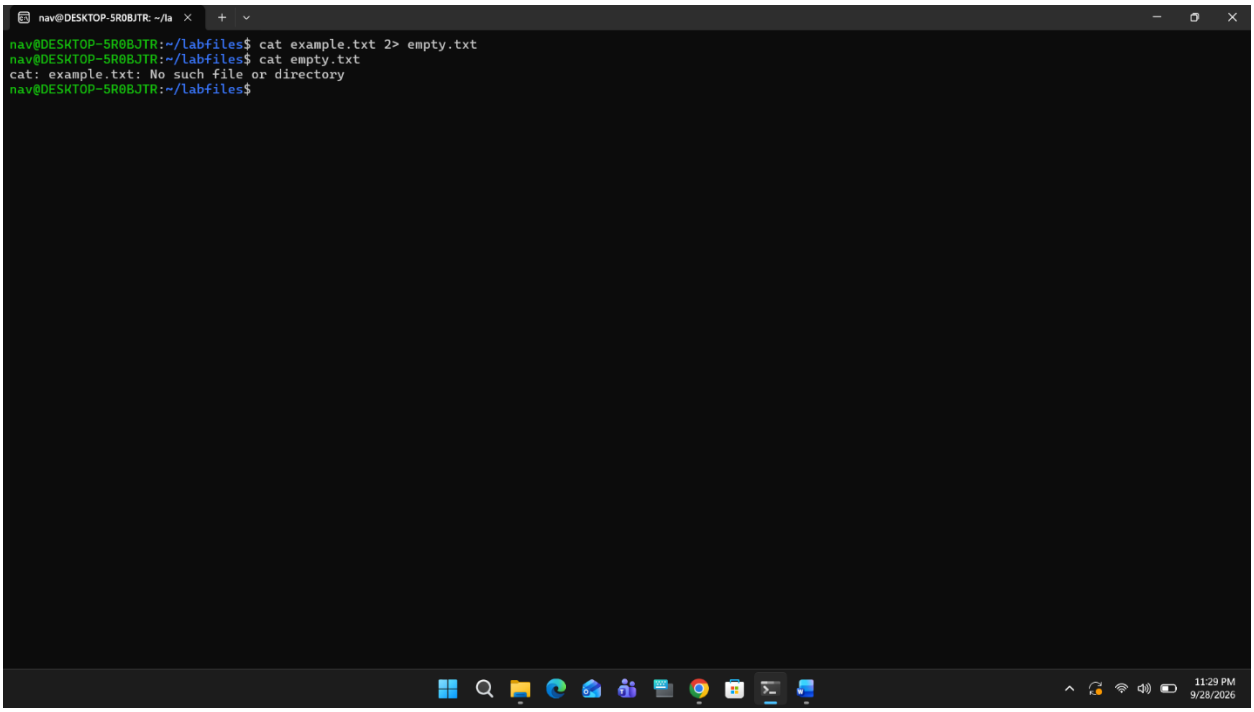
**Step 4 – Redirect standard input using <**

The < operator supplied students.txt as standard input to wc. This demonstrated input redirection and counted the lines in the file.



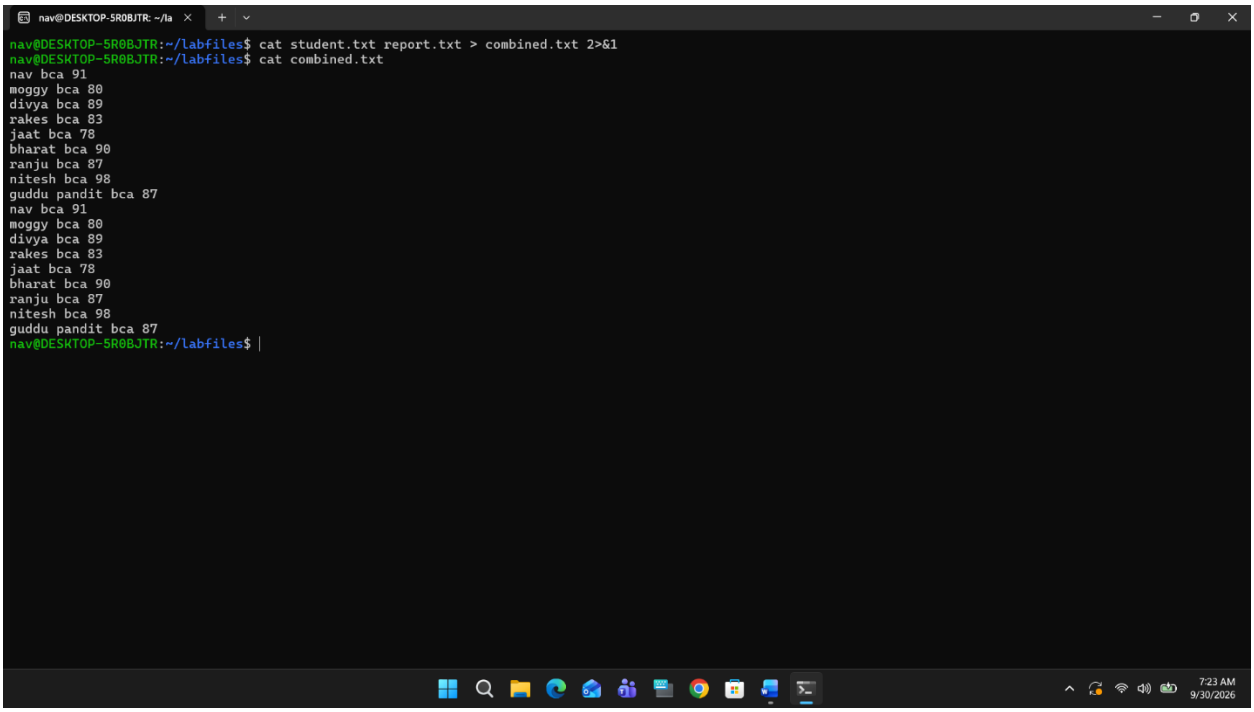
**Step 5 – Redirect errors using 2> and 2>>**

The 2> operator redirected an error message into a file. The 2>> operator is the append form of error redirection and is included with this step as requested.



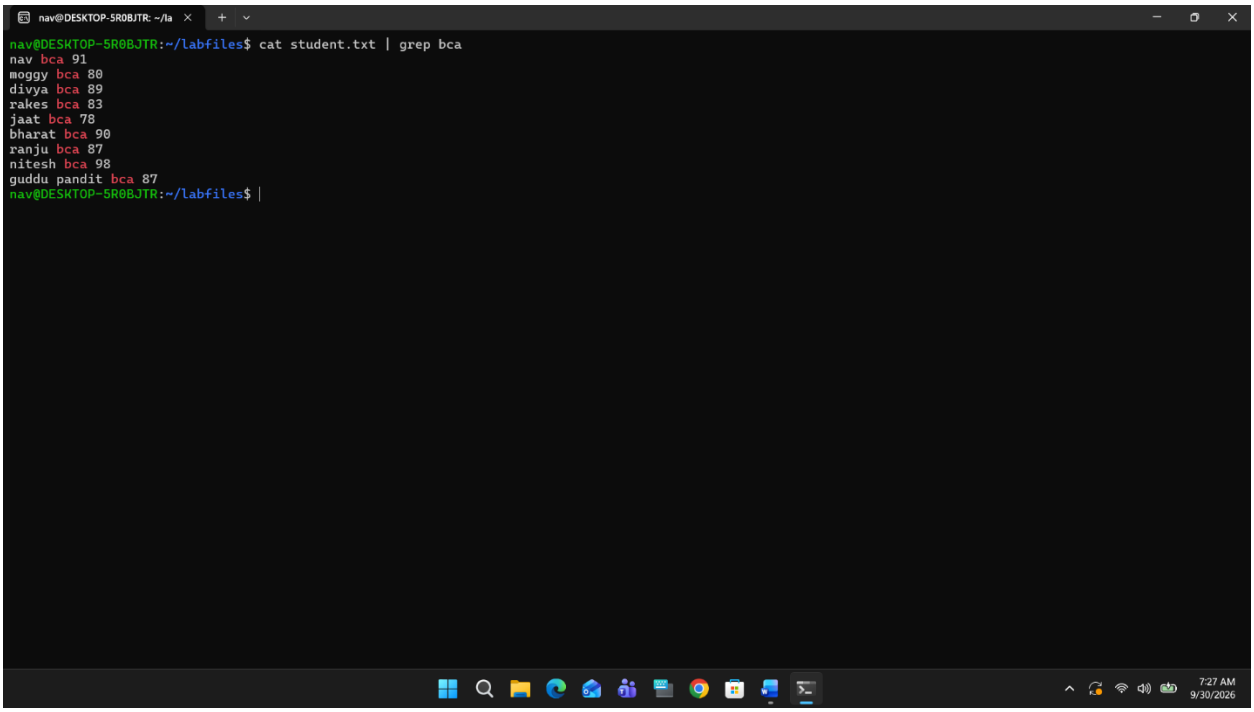
**Step 6 – Combine standard output and standard error using 2>&1**

The command demonstrated how standard output and standard error can be redirected to the same destination using 2>&1. The resulting file was then displayed for verification.



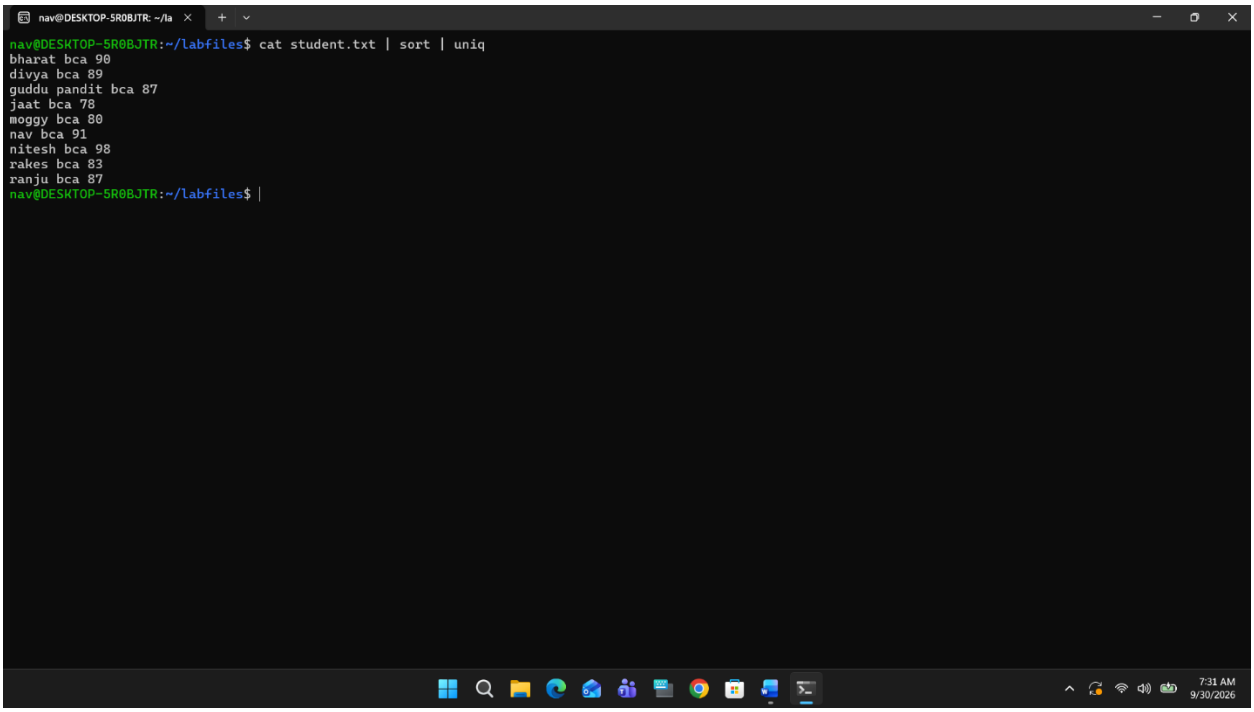
**Step 7 – Filter records using a pipe and grep**

The pipe operator | passed the output of one command directly to grep. grep filtered the student records and displayed only the entries containing MCA.



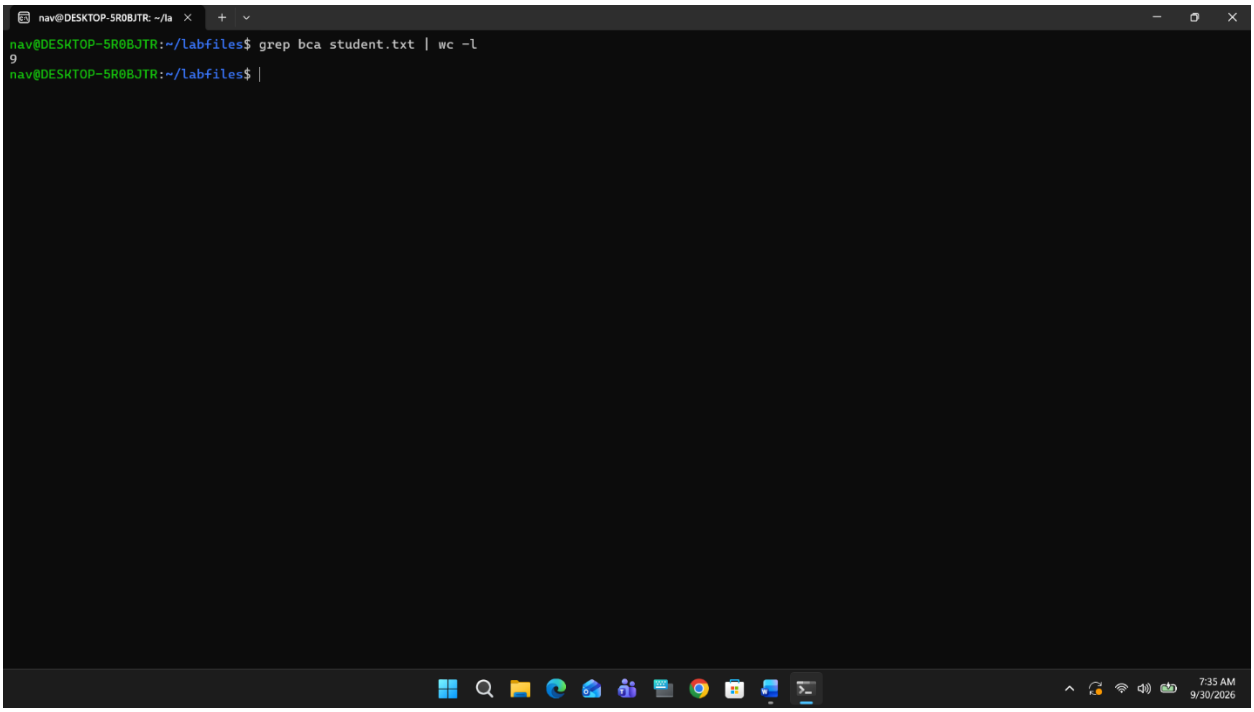
**Step 8 – Sort and remove duplicate lines using sort and uniq**

The output was passed through a pipeline. sort arranged the records, while uniq removed duplicate lines from the sorted result.



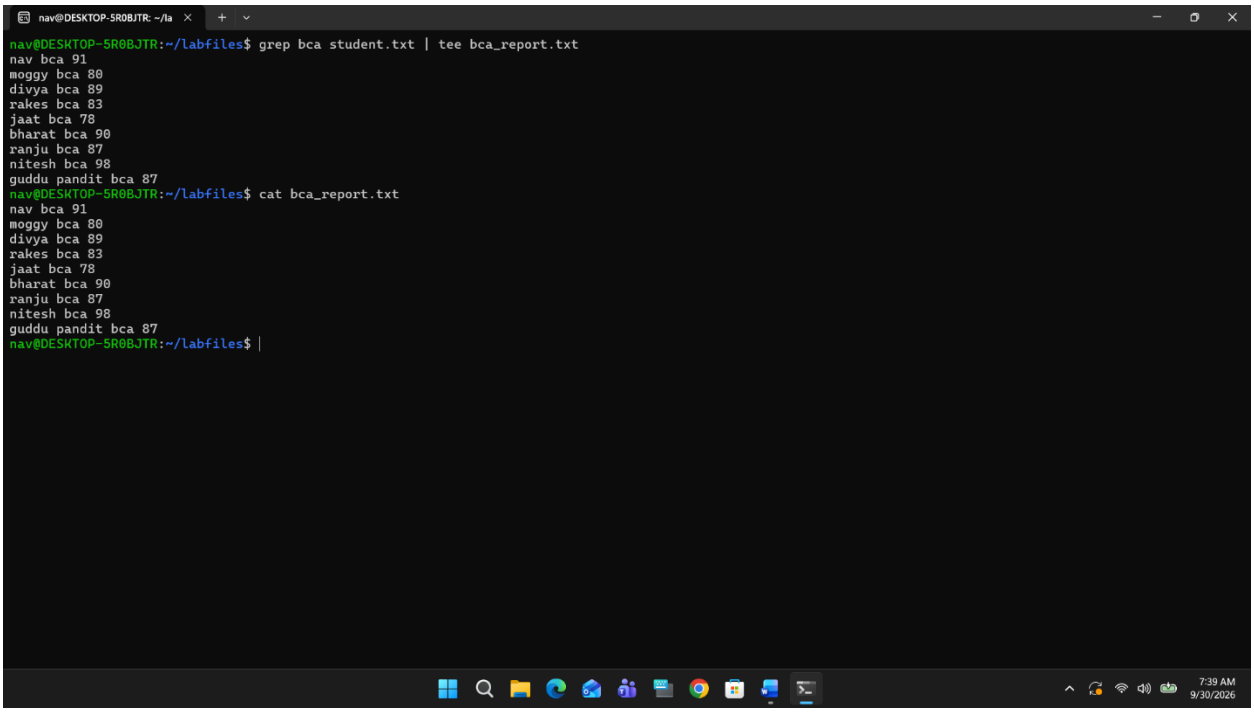
**Step 9 – Count filtered records using wc**

grep first selected the MCA records, and the pipe passed those results to wc -l. This demonstrated counting filtered output.



**Step 10 – Save filtered output using tee**

The tee command displayed the filtered MCA records on the terminal while simultaneously saving the same output into a report file.



**Step 11 – Generate the final filtered report**

The final pipeline combined cat, grep, sort, uniq, the pipe operator and tee to generate a filtered MCA report. The saved report was then displayed to verify the result.

